



Team ID	3			
Semester	7 th			
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Project Title	Natural Language Search Engine for Large-Scale Video Reposi-tories			
Centre of Excellence	Centre of Excellence in Computer Vision Research			
Internal Guide				
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INTRODUCTION:

Large video repositories such as surveillance archives, lecture libraries and media collections are indexed by nothing but filename and clock time, so footage can be retrieved only by when it was recorded, while what a user actually possesses is a **description** of the event. This project builds a **natural language search engine over video**: vision-language, speech and retrieval models derive a searchable description of footage that carries none, forming a **Retrieval-Augmented Generation (RAG) pipeline over video** that returns timestamped, playable moments and cited answers rather than a list of files.

OBJECTIVES:

1. To develop a pipeline that renders a recording searchable by description, without relying on shot boundaries, speech or visual salience.
2. To build a multi-vector semantic index returning ranked moments with **mm:ss** timecodes under typed filters.
3. To derive video-level understanding (summary, chapters, events, entities and anomalies) and answer questions against it with mandatory citations.
4. To deliver a multi-tenant web workspace with an LLM agent front end, over a self-describing HTTP API, and benchmark retrieval quality and latency at repository scale.

METHODOLOGY:

1. **Chunking:** Four fused boundary detectors: speaker change (pyannote), silence (Silero VAD), hard cuts (PySceneDetect) and semantic drift (OpenCLIP). A signal that produces no events cedes its weight to those that did
2. **Analysis:** Scene, person, object (YOLO11), on-screen text (EasyOCR) and speech (faster-whisper) analyzers, with the vision-language model gated by the cheap detectors and adaptive frame sampling

- 3. **Indexing:** Local BGE embeddings into Qdrant, five named vectors per chunk with a typed payload for structured filtering
- 4. **Aggregation & QA:** Twelve dependency-ordered aggregators; questions routed to the aggregate able to answer them, answers synthesised with cited moments
- 5. **Application:** FastAPI service and Next.js workspace with an agent tool loop; Supabase Postgres, Auth and Storage with row-level tenancy

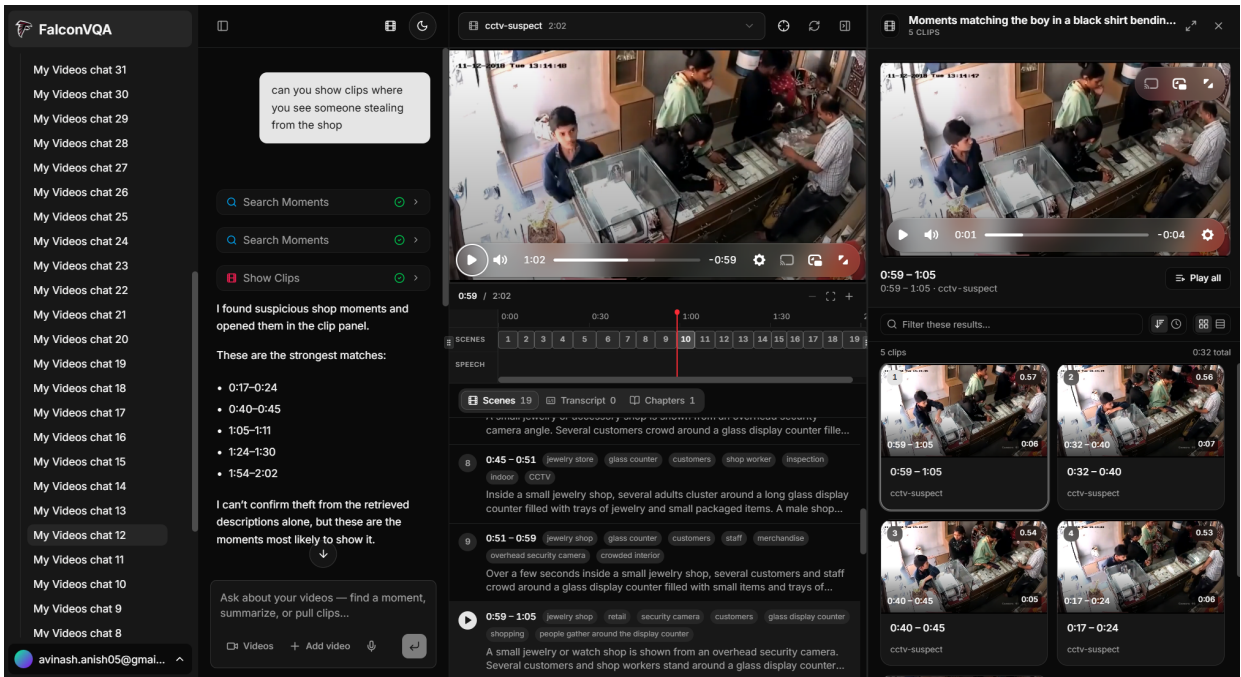
SOFTWARE REQUIREMENTS:

Python 3.13	FastAPI / Uvicorn	PyTorch (CUDA)	Transformers
Qdrant (vector DB)	sentence-transformers	faster-whisper	YOLO11 / EasyOCR
pyannote.audio	Silero VAD / PySceneDetect	OpenCLIP / OpenCV	Next.js / React / TypeScript
Supabase (Postgres, Auth, Storage)	Vercel AI SDK	Docker	Git

HARDWARE REQUIREMENTS:

Component	Specification	Component	Specification
Processor	Intel i5/i7 or equiv.	RAM	16 GB recommended
GPU	NVIDIA CUDA, 8 GB VRAM	Storage	50 GB (media cache + weights)

CONCEPT MOCKUP:



Concept mockup of the proposed workspace: a plain-language request resolved into ranked moments, with the retrieved clips and their m:ss timecodes shown alongside the answer.